

Stathis Galanakis

RESEARCH ENGINEER | MULTIMODAL GENERATIVE AI | MACHINE LEARNING

London, UK

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Summary

Research Engineer (PhD, Imperial College London) specialising in generative AI, diffusion models, and multimodal machine learning systems. My work focuses on scalable generative modeling under low-data and real-world constraints, with applications spanning synthetic data generation, representation learning, and identity-consistent generation. Published first-author research at ICCV and WACV, with experience building large-scale datasets, training pipelines, and generative ML systems across academia and industry.

Experience

Research Associate in Generative Computer Vision Models

London, UK

Department of Computing, Imperial College London

March 2025 - Present

- Developed rectified flow-based image synthesis for low-data augmentation, resulting in an 8% improvement in pathology classification.
- Scaled training of high-resolution generative models by deploying multi-GPU workflows (PyTorch DDP).
- Engineered LLM workflows for synthetic medical caption generation (500k image-text pairs) and explored automated pathology report classification.
- Supervised Master's students on research projects in generative AI and world modeling.

Research Engineer (Internship)

London, UK

Huawei UK

January 2022 - March 2025

- Built and evaluated diffusion-based generative models for robust identity-preserving image synthesis under ambiguous visual conditions, achieving sota identity preservation performance.
- Curated synthetic identity datasets (10k+ identities) to improve model robustness and generalisation.
- Designed neural rendering pipelines for controllable multi-view avatar generation and consistent viewpoint synthesis, across varying expressions, poses and lighting conditions.

Research Assistant

London, UK

Business School, Imperial College London

February 2021 - January 2022

- Applied machine learning to multimodal satellite imagery and weather data for crop yield prediction under spatial data sparsity.
- Developed synthetic data augmentation strategies to improve generalisation in underrepresented geographic regions.

Computer Vision Scientist

London, UK

ArielAI (acquired by Snap Inc.)

January 2020 - September 2020

- Built large-scale data collection and filtering pipelines for training vision models, incorporating human-in-the-loop annotation to improve dataset quality.
- Contributed to scalable data infrastructure supporting model development in a startup environment, later acquired by Snap Inc.

R&D, ML Engineer

Athens, Greece

Pobuca Ltd

May 2018 - January 2019

- Designed and deployed an end-to-end product detection system for retail shelf images (25+ classes), integrating semi-automated annotation and training workflows.

Technical Strengths

Programming:	Python, CUDA, C++, Linux, Git, Docker
Generative AI:	Diffusion Models, Video Diffusion, Rectified Flows, Transformers
3D & Vision:	3D Face Reconstruction, 3D Scene Reconstruction, NeRF, 3D Gaussian Splatting (3DGS)
Frameworks:	PyTorch, PyTorch Lightning, PyTorch3D, Accelerate
Systems:	Distributed Training, Multi-GPU Scaling, SLURM, Large-scale Data Pipelines

Selected Publications

DERMAFLUX: Synthetic Skin Lesion Generation with Rectified Flows for Enhanced Image Classification Galanakis S , Koliouisis A, and Zafeiriou S.	<i>Under peer review</i>
DERMAFLUX is a rectified flow-based text-to-image model for synthetic skin lesion image synthesis.	
STARCaster: Spatio-Temporal AutoRegressive Video Diffusion for Identity- and View-Aware Talking Portraits Paraperas F, Galanakis S , Potamias R, Kainz B, and Zafeiriou S.	ICML 2026
SpinMeRound: Consistent Multi-View Identity Generation Using Diffusion Models Galanakis S , Lattas A, Moschoglou S, Kainz B, and Zafeiriou S.	ICCV 2025
SpinMeRound introduces a diffusion framework for identity-consistent multi-view visual generation from sparse inputs.	
ImHead: A Large-scale Implicit Morphable Model for Localized Head Modeling Potamias R, Galanakis S , Deng J, Papaioannou A, and Zafeiriou S.	ICCV 2025
FitDiff: Robust monocular 3D facial shape and reflectance estimation using Diffusion Models Galanakis S , Lattas A, Moschoglou S, and Zafeiriou S.	WACV 2025
FitDiff introduces a multimodal diffusion framework for robust identity-preserving visual generation from unconstrained images.	
3DMM-RF: Convolutional Radiance Fields for 3D Face Modeling Galanakis S , Gecer B, Lattas A, and Zafeiriou S.	WACV 2023
3DMM-RF explores neural radiance field representations for controllable and identity-consistent visual generation, under diverse expressions, poses and lighting conditions.	

Education

PhD in Computer Science Supervisor: <i>Prof. Stefanos Zafeiriou</i>	Imperial College London, UK April 2025
<i>Thesis:</i> Advancements in Face Reconstruction from a Single Image. Research focused on diffusion-based generative modeling and robust visual generation under unconstrained real-world conditions. Work resulted in peer-reviewed publications (ICCV, WACV), including first-author contributions.	
Diploma/M.Eng. in Electrical and Computer Engineering Supervisor: <i>Prof. Petros Maragos</i>	NTUA, Greece November 2019
<i>Thesis:</i> Human Action Recognition and Localisation in Videos.	

Interests

- Reviewer for international conferences such as CVPR (2024, 2025), ECCV (2026) and MICCAI (2026).
- Passionate about basketball, cycling, and running.